Experiment - 06

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A)

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

Sales

sale_date	fruit	sold_num
2020-05-01	apples	10
2020-05-01	oranges	8
2020-05-02	apples	15
2020-05-02	oranges	15
2020-05-03	apples	20
2020-05-03	oranges	0
2020-05-04	apples	15
2020-05-04	oranges	16

Output

sale_date	diff
2020-05-01	2
2020-05-02	0
2020-05-03	20
2020-05-04	-1

SOLUTION :

```
SELECT
    SALE_DATE,
    SUM(
```

```
    CASE
        WHEN FRUIT='apples' THEN SOLD_NUM
        ELSE -SOLD_NUM END) AS DIFF
FROM SALES
GROUP BY SALE_DATE
ORDER BY SALE_DATE ASC;
```

B)

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:
id represents the current node.
p_id represents the parent node.
Classify every node into one of the following categories:
Root: A node that does not have any parent.
Leaf: A node that has a parent but does not have any children.
Inner: A node that has both a parent and at least one child.
Display the node ID along with its category. Return the result sorted by node ID.

Tree

id	p_id
1	NULL
2	1
3	1
4	2
5	2

Expected Output

id	Type
1	Root
2	Inner
3	Leaf
4	Leaf
5	Leaf

Solution:

```
SELECT
    ID,
    CASE
        WHEN P_ID IS NULL THEN 'ROOT'
        WHEN ID NOT IN (
            SELECT
                P_ID
             FROM TREE
```

```
                  WHERE P_ID IS NOT NULL) THEN 'LEAF'
    ELSE 'INNER' END AS "TYPE"
FROM TREE;
```